

Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

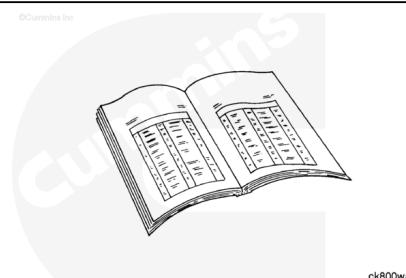
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



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- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html>)
- Remove the oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html>)
- Remove the turbocharger oil supply line from the oil filter head. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-046-tr.html>)
- Remove the turbocharger oil drain from the turbocharger. Refer to Procedure 010-045 in Section 10.

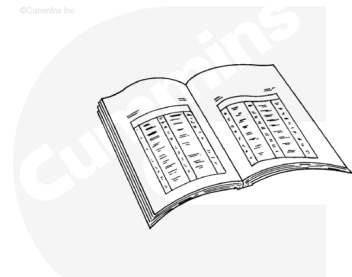
(/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)

- Remove the two support brackets from the filter head.

Automotive with CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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⚠ WARNING ⚠

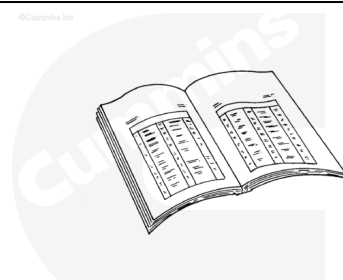
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)
- Disconnect the turbocharger control valve air line from the turbocharger control valve. Refer to Procedure 010-118 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-118-tr.html)
- Remove the turbocharger air control valve. Refer to Procedure 010-088 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-088-tr.html)
- Remove the oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html)
- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)
- Disconnect the engine brake oil supply line from the lubricating oil filter head.

Automotive with CM876

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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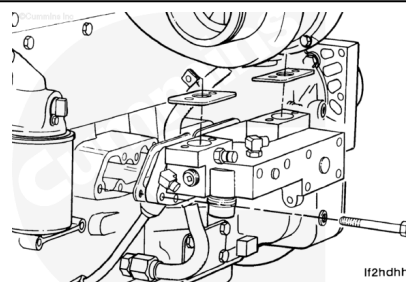
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html>)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html>)
- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html>)
- Disconnect the engine brake oil supply line from the lubricating oil filter head.

Remove

Remove the nine filter head capscrews and the filter head.
Remove and discard the gaskets.

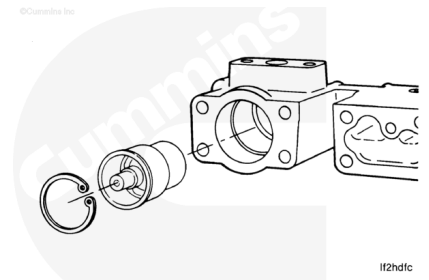


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Disassemble

Use snap ring pliers to remove the snap ring in the thermostat bore.

Remove the lubricating oil thermostat.



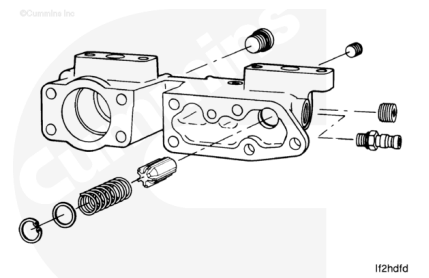
⚠ WARNING ⚠

The valve spring must be compressed to remove the snap ring. To avoid personal injury, wear face and eye protection.

Remove the snap ring from the bypass valve bore.

Remove the washer, spring, and bypass valve.

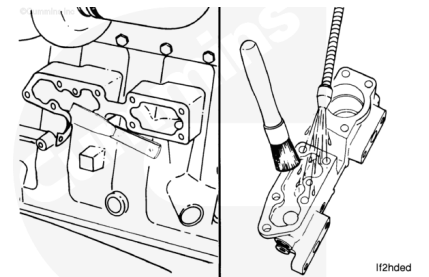
Remove the remaining plugs and fittings from the filter head.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

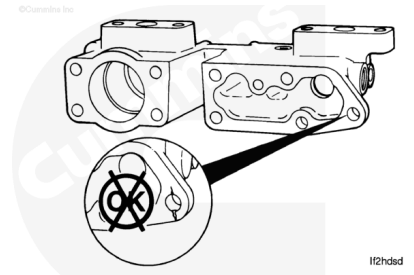
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the filter head and cylinder block gasket surfaces.

Use solvent to clean the filter head. Dry with compressed air.

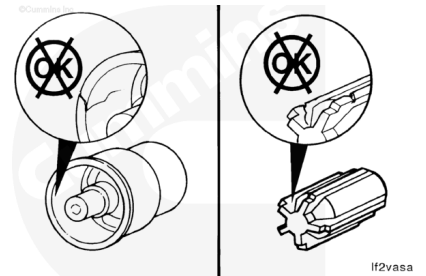
Inspect for cracks or other damage.

Inspect the lubricating oil thermostat. Refer to Procedure 007-039 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-039-tr.html>)



Inspect the thermostat and bypass valve for cracks or other damage.

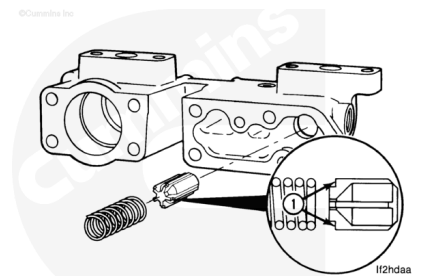
If the bypass valve or thermostat are damaged, they **must** be replaced.



Assemble

Install the bypass valve into the filter head. The notched end of the valve (1) **must** face the engine side of the filter head.

Insert the spring into the bore. Check to be make sure the spring is properly located over the notched end of the valve (1).

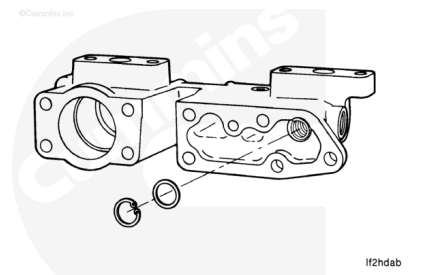


⚠ WARNING ⚠

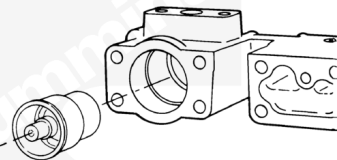
The valve spring must be compressed to install the snap ring. To avoid personal injury, wear face and eye protection.

Place the washer on top of the spring.

Use snap ring pliers to install the snap ring into the groove inside the bore.



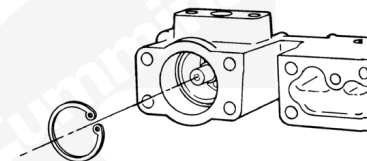
Insert the oil thermostat into the filter head. The beveled end of the thermostat is inserted first.



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Use snap ring pliers to install the snap ring into the grooves.

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Install the following fittings into the filter head.

Torque Value:

Turbocharger oil supply fitting (1) 12 n•m [106 in-lb]

Torque Value:

Engine brake supply fitting (2) 12 n•m [106 in-lb]

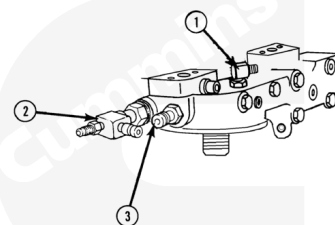
Torque Value:

Compuchek™ fitting (3) 25 n•m [221 in-lb]

Torque Value:

Remaining plugs 25 n•m [221 in-lb]

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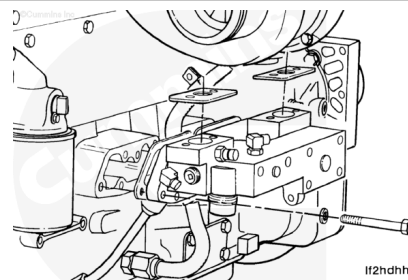
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Install

Use a new gasket and install the filter head to the cylinder block.

Install the nine capscrews and tighten.

Torque Value: 47 n•m [35 ft-lb]



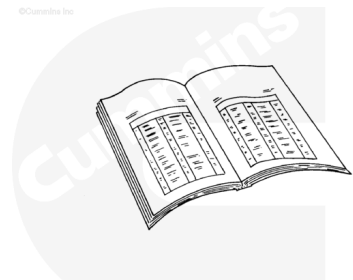
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Finishing Steps

Automotive with CM875, CM870 and CM570

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



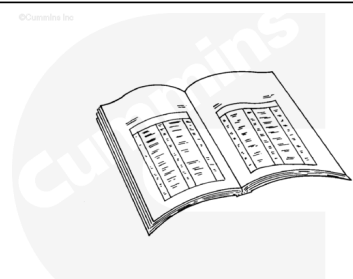
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- Install the lubricating oil cooler. Refer to Procedure 007-003 in Section 7.
(/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)
- Install the turbocharger support brackets.
- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-046-tr.html)
- Install the turbocharger oil drain lines. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html)
- Filling the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine at idle speed for 3 minutes to inspect for leaks at the filter head.
- Confirm the correct oil level in the oil pan.

Automotive with CM570

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to reduce the possibility of engine damage.



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- Install the lubricating oil cooler. Refer to Procedure 007-003 in Section 7.

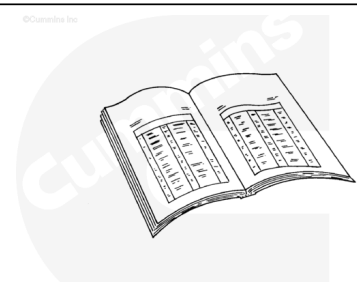
(/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)

- Connect the engine brake oil supply line to the lubricating oil filter head.
- Install the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)
- Install the oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html)
- Install the turbocharger air control valve. Refer to Procedure 010-088 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-088-tr.html)
- Connect the turbocharger control valve air line from the turbocharger control valve. Refer to Procedure 010-118 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-118-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine at idle speed for 3 minutes to inspect for leaks at the filter head.
- Confirm the correct oil level in the oil pan.

Automotive with CM876

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to reduce the possibility of engine damage.



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- Install lubricating oil cooler. Refer to Procedure 007-003 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-003-tr.html)
- Connect the engine brake oil supply line to the lubricating oil filter head.
- Install the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)

- Install the oil filter. Refer to Procedure Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/35/35-007-013-tr.html)
- Fill cooling system. Refer to Procedure Refer to Procedure 008-018 in Section 8.
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- Operate the engine at idle speed for three minutes to inspect for leaks at the filter head.